

Name: _____

Answer all questions, assuming I start with the following code :

- Assume the producer runs :

```
sem_wait(&empty);  
put(val);  
sem_post(&full);
```

- And the consumer runs :

```
sem_wait(&full);  
val = get();  
sem_post(&empty);
```

1. Assume the buffer holds 10 values. What should each semaphore be initialized to?

2. Explain a situation where this code could deadlock.

3. How would you fix the deadlock? Explain where in the code you would put this fix.